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Technology Solution Drilling Through Extremely Abrasive Heterogeneous Tight Sandstones

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Abstract

Drilling hard abrasive rock has been very problematic for oil and gas companies. Sandstone and Siltstone layers, with high Unconfined Compressive Strength (UCS) values, are drilled in single or interbedded layers sections to reach the casing point (CP). One of the challenges is the number of bits needed to reach CP. Significant cost is observed due to the additional days needed to drill the wells. This comes from the tripping time exerted due to early bit wear before reaching target depth (TD). The objective was to introduce a new product while applying efficient drilling practices to increase the drilling interval with relatively high ROP and reduce drilling operations cost.

Multiple wells have been drilled in different areas. These wells drilled through hard sandstone and siltstone lithology. The common belief in the industry is to use PDC bit with continuous shearing mechanism. The theory states that the rotation mechanism of the PDC cutters ensures that the bit stays sharp for a longer time which allows it to drill further in the sandstone environment. High cost has been encountered by deploying 4 to 8 PDC bits per well to reach the section TD.

Hole problems such as under-gauge section, bit junk in the well, twisted off/parted Bottom Hole Assembly (BHA) are major issues in this application. Vertical mapping of high UCS stingers for ongoing sections is difficult due to deposition sequence. Ensuring optimum parameters implementation, performing assessments for pre-run, ongoing run and post run play essential role to benefit from this hybrid approach.

The problem was studies and a new product that combines roller cone and PDC cutting elements was introduced in the same application. Further analysis has been conducted to validate the benefit of the hybrid design. UCS calculations as well as lithology percentage were analyzed to understand the added value of the product introduction.

The product was deployed in multiple areas to study the bit behavior. In area of interest A, 4 bits were consumed per well. The new product was able to reduce the number of consumed bits from 4 to 2. In area of interest B, bits were averaging 200 ft per run. When the new product was deployed, the interval per bit increased to 800 feet, which improved the efficiency by 275%. UCS as well as siltstone percentage was compared to have an apple-to-apple comparison. The data shows higher ROP as well as longer footage achieved while drilling the same siltstone percentage and using the same parameters.

Combining the PDC efficient shearing of the formation with the advantage of the roller cone crushing action allows to drill faster and longer intervals while applying efficient parameters. This paper highlights how the hybrid approach can contribute to significant cost reduction, time saving and necessary avoidance of side-track jobs.

Introduction

One of the main challenges of seeking hydrocarbon is drilling through hard and abrasive lithology. In many cases, multiple bits would be needed to finish long sections of hard and abrasive sandstone. This adds cost due to the number of drill bits utilized and the tripping time. Rig rate plays a major role in drilling costs and every hour saved is a tremendous cost reduction in the drilling operation.

Hard rock drilling has been a major problem since Oil and Gas companies started seeking gas in deep layers of the earth. These layers normally consist of sandstone and siltstone. The abrasive nature of these lithology leads to drill bit accelerated wear and added costs to the drilling operation. This adds cost to the drilling operation and leads to reduction in the economic value to seek energy in these territories. Development in drill bit technologies for hard rock drilling date back to the 1900s when roller cone bits were first used. In 1909, Howard R. Hughes revolutionized drilling by inventing the two-cone rotary drill bit. It allowed more efficient drilling through hard rock formation compared to the traditional fishtail bits, (ASME, 2025).

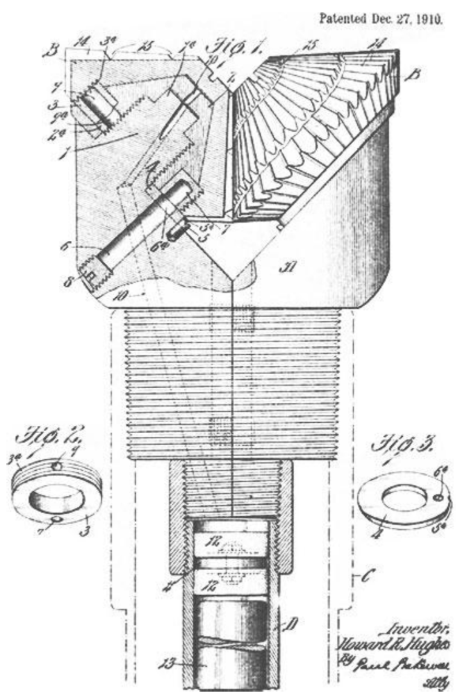

Figure 1—Hughes Two-Cone Drill Bit Patent Drawing

Later the Polycrystalline Diamond Compact (PDC) Bits were introduced in the 1980s with synthetic diamond cutters. The diamond cutters increase durability and wear resistance compared to roller cone bits and allowed drilling further in hard lithology (Natioanl Renewable Energy Laboratory, 1998). Further developments in PDC cutters wear and impact resistance have improved drilling in hard rock in the 2000s and later in the 2010s with improved diamond grades and diamond leaching technology (Green, 2002), (Scott, Meiners, & Isbell, 2012).

Figure 2—PDC Drill Bit Dull Picture

In the recent years in 2013, further innovative solutions were introduced including using a rolling PDC cutter that revolves around its own disc center to use the whole 360 circumference of the cutter. This has been a breakthrough in drilling technology, and it allowed drilling even further in hard and compacted sandstone environment (Zhang, et al., 2013). This has been the main product to use in hard and compacted sandstone drilling environment until today. However, although it has its benefits of increasing diamond available to wear down while drilling the abrasive sandstone, it also has its downside as these rolling cutters could break and lead to under-gage bits.

Figure 3—Rolling Cutter on Left. Bit with Broken Cutters on Right

During the focus on PDC cutters technology, other technology leads looked at a different approach of increasing drilling efficiency. Development was focused on hybrid drill bit designs that combine the shearing action of the PDC cutters as well as the crushing action from the roller cones. These hybrid designs were introduced in the early 2010s but were primarily focused on big diameter bits that have interbedded sections of hard and soft lithology. These designs made waves and improved drilling efficiency, but they were not targeted at hard and compacted smaller diameter holes that drill through hard and abrasive siltstone and sandstone. In the Gulf of Mexico, an oil rig improved drilling performance by utilizing a 26″ hybrid design. Roller Cone and PDC bits were previously used. After deploying the hybrid design, 50% to 137% improvement in ROP was seen (Chowdhury, et al., 2017).

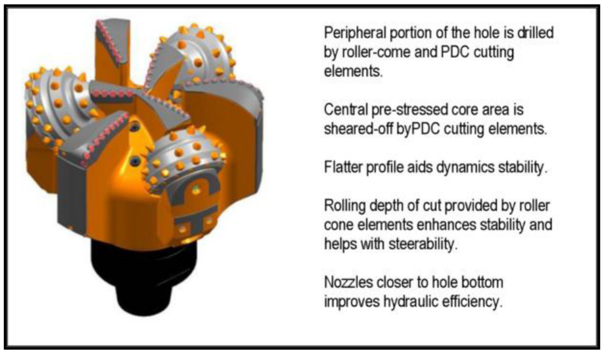

Figure 4—Hybrid Drill Bit Design That Combine PDC Cutters and Roller Cones

Additionally, due to the stigma associated with fallen roller cones, these designs were disregarded as an option in some areas where they drill hard and abrasive sandstone. However, in recent years with the improved retention mechanism for these cones, as well as in the design of the bit, the hybrid drill bits showed further improvement in hard rock drilling. Some of these retention mechanism techniques has been patented by the United States (USA Patent No. 9,004,198, 2015). This paper will be highlighting how hybrid designs excelled at increasing interval drilled in hard and interbedded lithology compared to the designs with rolling cutters.

Methodology

The 8.375″ Hybrid Drill Bit design was introduced in two different areas. In both areas the same lithology, consisting of hard, compacted sandstone interbedded with siltstone is drilled. In area of interest A the formation is drilled vertically on Motor (M/MTR) or Rotary Steerable System (RSS) or MRSS to create a slight deviation from vertical. The drill bit consumption can reach four to eight bits per well. The top section of the well is around 3,500 ft of shale and siltstone and is normally easy to drill with one bit compared to the lower part of the section.

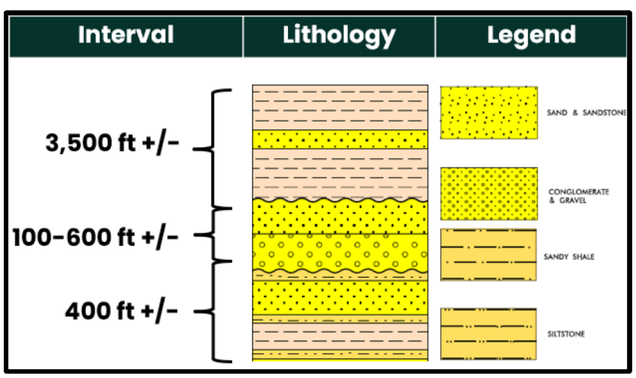

Figure 5—Area of Interest A Typical Lithology Drilled

The Rate of Penetration (ROP) in the top part can reach 30 to 80 ft/hr on average. Typically, Fixed Cutter PDC and/or Rolling Cutter Drill Bits are used, in which they are usually efficient in this soft lithology.

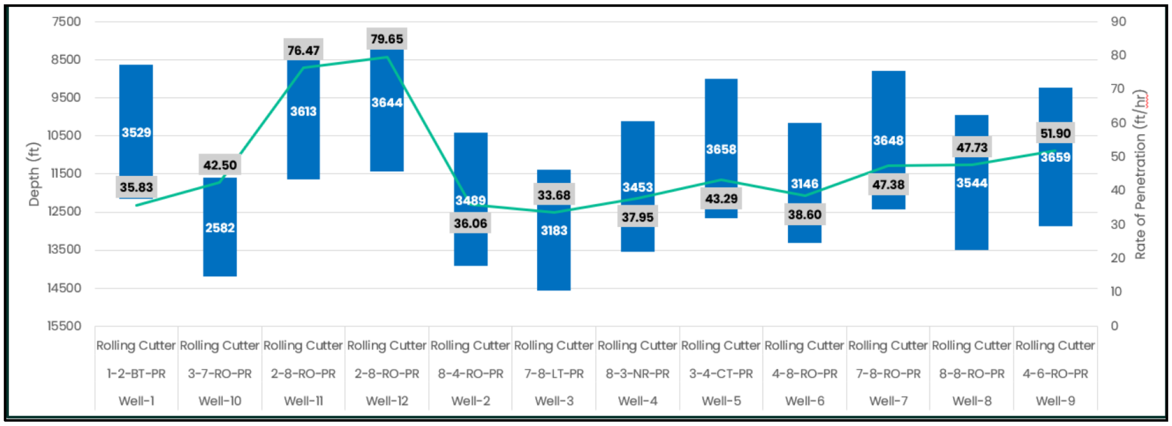

Figure 6—Area of Interest A - Top Section Benchmark

However, once the bit reaches the lower compacted sandstone of the section, ROP can significantly drop to an average of 4 to 8 ft/hr. In most of the wells, the bit would not be able to drill through this tough sandstone lithology and the BHA would be switched to Rotary System (RTR). An extra 3 to 7 bits would be needed to reach the section total depth (TD). Multiple issues arise in this section including bit debris (junk) left in the well, under-gauge holes, increased torque, twist-offs or parting of BHA. These complications often arise due to the complex interaction between the bit and heterogeneous formations, particularly in intervals with high Unconfined Compressive Strength (UCS) features such as stingers.

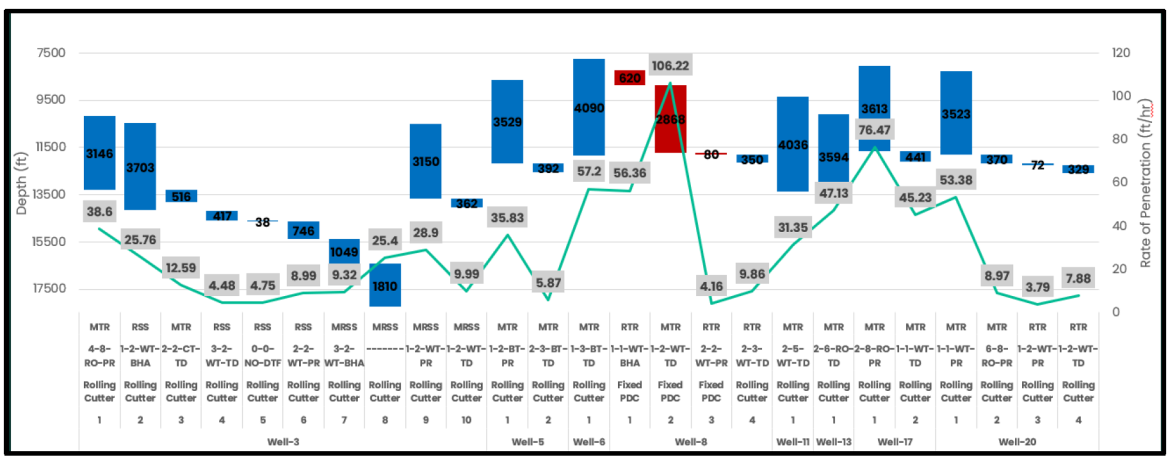

Figure 7—Area of Interest A - Lower Section Benchmark

The last part of the section can be relatively easy when the lithology shifts back to claystone again.

On the other hand, Area of Interest B is a lateral section through hard and abrasive sandstone interbedded with siltstone. It is drilled laterally with RSS system. The total lateral interval is around 4000 to 5000 ft. Sometimes the BHA would be switched to MRSS to increase drilling efficiency when the well becomes too difficult to drill. The number of bits used to finish these laterals varies from well to another. Each well may utilize 4 to 10 bits to reach TD with ROP that ranges from 5 to 30 ft/hr. Each bit could drill 200 ft to 2,000 ft depending on how difficult the well is.

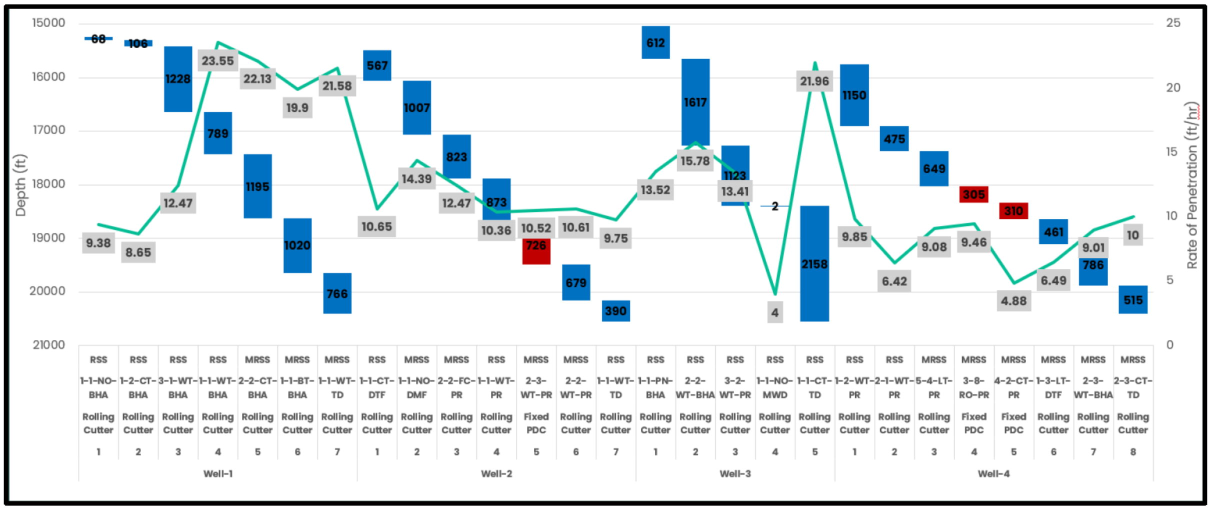

Figure 8—Area of Interest B - Lateral Section

One of the most complex aspects of drilling in these areas is conducting vertical mapping of high UCS stingers. These are localized, extremely hard zones that are difficult to predict due to the irregular and layered nature of sedimentary deposition. These stingers can cause mechanical damage, leading to cutter breakage, inefficient penetration, or even catastrophic BHA failures if not properly anticipated.

To mitigate these risks and optimize drilling performance, it is essential to implement a structured approach to parameter management and operational assessment. This includes:

- **Pre-Run Planning**: Conducting detailed formation analysis using offset well data, UCS modeling, and mechanical property logs to anticipate drilling hazards and select the most suitable bit and BHA configuration.
- **Ongoing Run Monitoring**: Real-time tracking of drilling parameters such as torque, WOB, ROP, and vibration. This enables proactive adjustments to mitigate shock, stick-slip, and cutter overload —especially when transitioning into harder zones.
- **Post-Run Evaluation**: Analyzing bit dull condition, identifying wear mechanisms, and correlating performance with formation characteristics. This feedback is critical for continuous improvement in bit selection and drilling strategy.

Despite these efforts, the limitations of conventional bit designs in such environments often result in poor performance and increased non-productive time (NPT). In both areas, the primary drill bit used is the rolling cutter. Although good results have been seen with this type of drill bits compared to the conventional fixed cutter PDC designs, the performance has not always been consistent. Especially when drilling the challenging wells where the formation becomes very interbedded between hard sandstone and siltstone. As a result, this leads to under-gage holes, short trips and low rate of penetration.

These challenges set the stage for introducing hybrid bit technology as a potential solution—offering a more robust and adaptive cutting structure capable of handling the variability and toughness of hard sandstone formations. To save cost and improve performance the hybrid concept was utilized. The hybrid design was deployed in three wells of areas of interest A with total units of fours. In Area of Interest B, the hybrid design was deployed in one well with a total of 3 units. Further analysis was done to validate the performance and assess if the technology adds value to the drilling operation. UCS values for the wells drilled were compared and lithology percentages were analyzed to reach a more accurate conclusion.

Results and Observations

Area of Interest A

First Deployment. In the first well of Area of Interest A, the hybrid design was deployed due to the struggle to reach TD. The PDC bit with rolling cutters was used from the beginning of the section until it drilled 129 ft inside the hard sandstone formation. The bit was pulled out of hole for low penetration rate. Two other bits with rolling cutters were used on motor and rotary BHA but each drilled low footage of 31 and 41 ft and pulled out for penetration rate. The decision was made to run a hybrid design to investigate alternative solutions. Pre-run analysis as well as 24-hour monitoring of bit behavior and parameters optimization based on bit response was conducted. Particular attention was paid to pull off bottom when ROP drops and reapply weight on bit (WOB) rather than stacking weight as a practice. ROP limit was applied to around 25 ft/hr instantaneous to mitigate impact damage that could happen while drilling through hard and soft layers. These practices were followed during all these runs.

As can be seen from Figure 9, the Hybrid Drill Bit was able to drill the remaining 123 ft of hard sand and continue to section TD in one run. The bit drilled a total of 414 ft with 9.57 ft/hr and managed to come out in worn cutters rather than broken cutters. Post run evaluation was conducted to understand the actual environment each bit went through. As can be seen from Figure 10 from the UCS column, the first Bit with Rolling Cutters drilled sand that had UCS values of around 28 ksi and less. Lithology was slightly interbedded, but the bit only drilled 2 stingers which caused mechanical breakage to cutters.

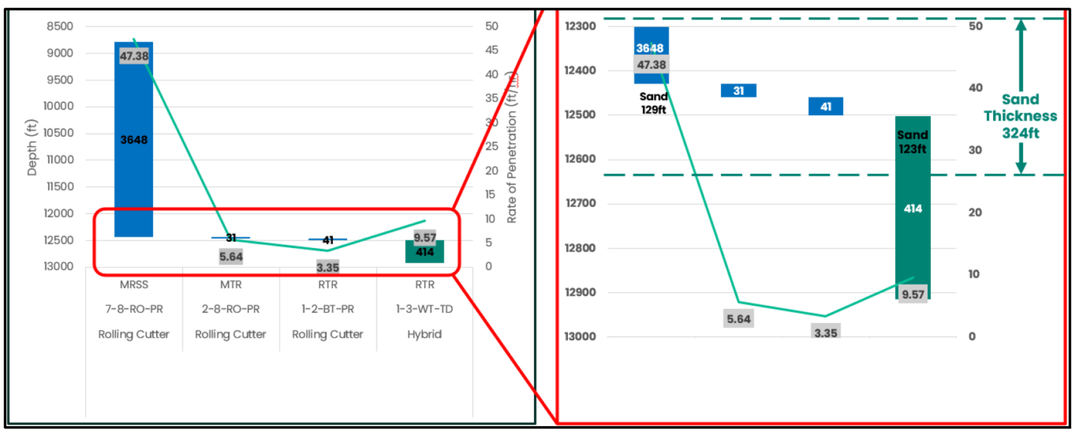

Figure 9—Area of Interest A - First Deployment of Hybrid Drill Bit

Figure 10—Dull Pictures of Hybrid Bit After Drilling 4 Stingers

The second bit with rolling cutters looked like it was not able to sustain one stringer that reached 35 ksi. The 3rd bit deployed drilled at a more homogenous UCS values with one stinger reaching 35 ksi but was still pulled out for penetration rate and came out with some broken cutters. On the other hand, compared to the previous runs, the Hybrid Bit drilled harder environment. It drilled at the same UCS with more interbedded stingers. More siltstone was present between the stringer which makes the chance of hitting these stringers higher. The hybrid design was able to survive 4 stingers with UCS above 30 ksi and finish to section TD.

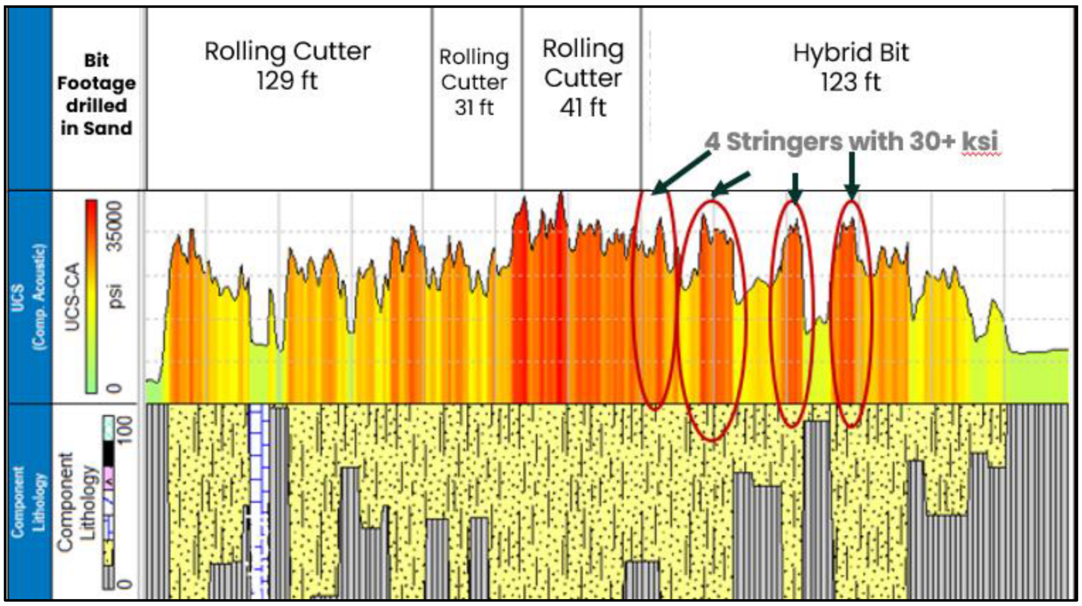

Figure 11—Area of Interest A – First Deployment UCS Calculations

Second Deployment. In the second well, a conventional fixed cutter bit was utilized before reaching the anticipated depth, when the hard sandstone would be encountered. Then, the decision was made to go with the hybrid design on RSS BHA to deviate slightly from vertical then drop again as per directional plan requirement. The first hybrid design drilled 734 ft out of which is 338 ft in the abrasive sand lithology. The bit was pulled out of hole to switch from RSS to Rotary BHA and was found with some broken cutters. However, the bit was in-gage, and the subsequent run did not encounter any obstructions, nor did it need to ream down the hole. The second hybrid design was made up on rotary BHA and continued to drill to section TD. The bit drilled 564 feet out of which is 314 feet inside the abrasive sand lithology. The two bits drilled the longest abrasive sandstone interval in this area of around 652 ft when normally 4 to 8 bits are needed. Not only cost was saved by utilizing less bits, but tremendous time was also saved as less trips needed to switch bits.

As can be seen from Figure 12, the two Hybrid bits drilled at UCS values that ranged from 20 to 28 ksi. Although there were some stingers in this well, they were not as severe compared to the first well. The UCS values in this well look more homogenous and forgiving. However, compared to the first run in well-1 with the rolling cutters, these two bits drilled at the same UCS values with very similar UCS range and stinger values. Yet, the rolling cutter came out with very sever dull and the hybrid bits came out in-gage. This saves NPT cost due to reaming the hole before starting to drill again.

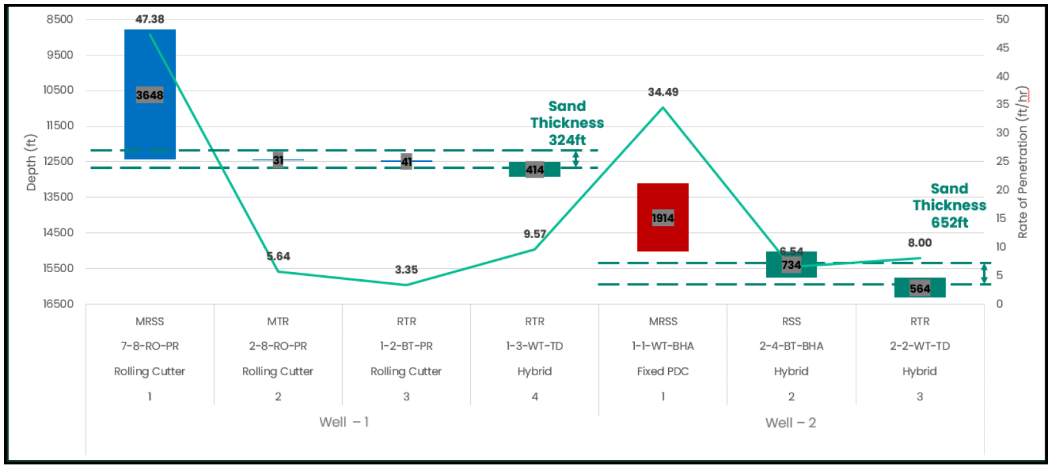

Figure 12—Area of Interest A - Second Deployment of Hybrid Drill Bit

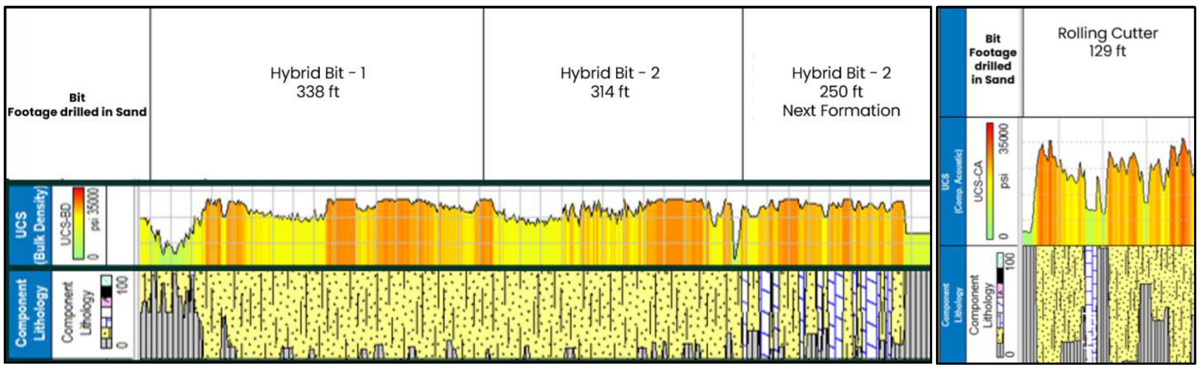

Figure 13—Area of Interest A – Left: Second Deployment UCS Calculations – Right: First Rolling Cutter Bit in Well-1

Third Deployment. In the third well, the Rolling Cutters-bit was used to drill until the ROP dropped below acceptable levels. The bit came out with very sever dull conditions and it was found 2″ under-gage after drilling 173 ft in the sand lithology. This means the next bit will need to ream down however interval and enlarge it the normal size. With 2″ under-gage hole, the outer portion of the cutting structure of the next bit will have high loading on a concentrated area compared to normal drilling. The decision was made to pick up the hybrid bit next. The drill bit had to ream down an estimated 500 to 700 ft at various hole gage diameter before tagging bottom and starting to drill. The bit drilled 58 feet in sand lithology and was pulled out of hole for low penetration rate. The bit was found with worn cutters rather than broken cutters which is more desirable as it means no junk or debris are left in the hole. The next bit that was picked up was the rolling cutter bit which drilled 98 ft and was pulled out for low ROP. The bit was found with severely broken cutters. A fourth bit was needed to finish the section, and it was made up on motor BHA. The bit drilled 170 ft of sand and continued until section TD. The bit was found with broken cutters.

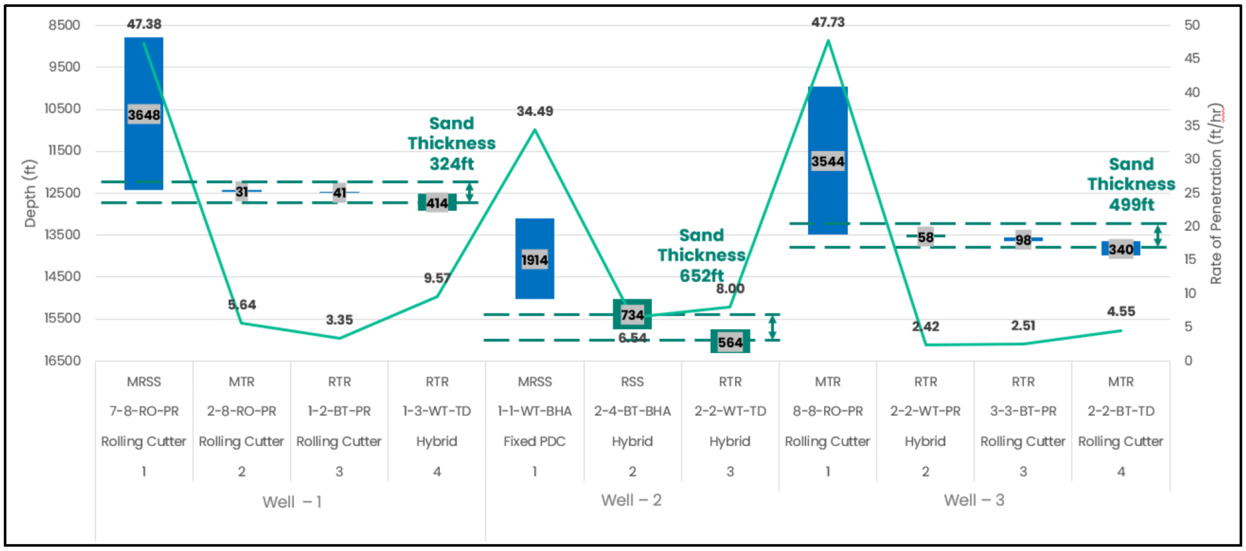

Figure 14—Area of Interest A - Third Deployment of Hybrid Bit

Although, at first glance, it looked like the hybrid bit drilled the shortest interval, a couple of points need to be emphasized. The bit reamed down at least 500 feet of abrasive sand before drilling. The bit was pulled out of hole with normal wear and in-gage. Upon looking at the UCS calculations, the hybrid design drilled the hardest interval in the well with stingers reaching 38 ksi and yet coming with normal wear. Although ROP increased after picking up the subsequent Rolling Cutter bit, ROP also dropped again to the same level when UCS increased. This means that the hybrid bit could have continued drilling further if operations were more patient.

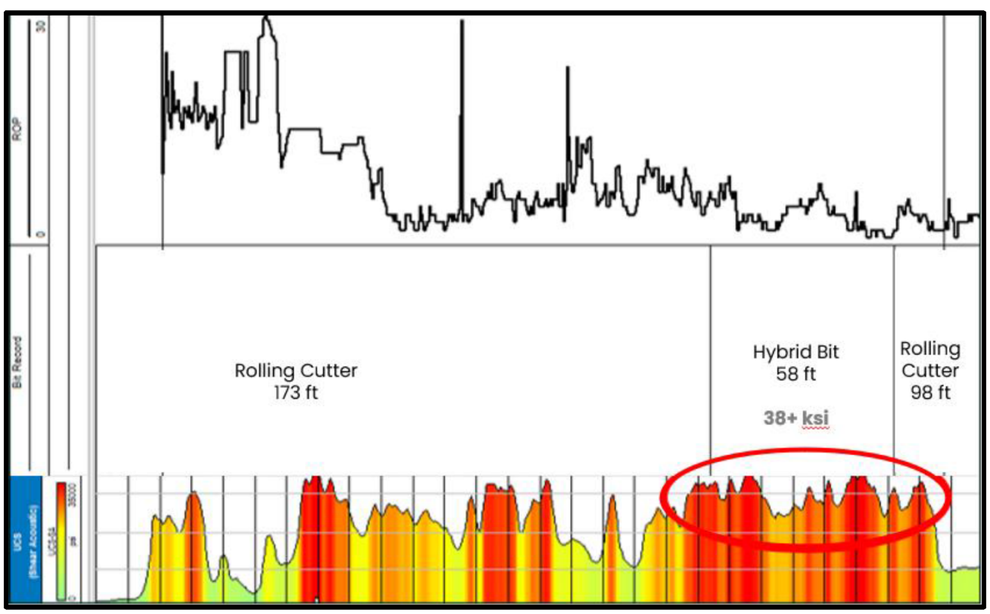

Figure 15—Area of Interest A – Third Deployment UCS Calculations

In summary, the results of Area of Interest A deployment show consistently better dull conditions of the hybrid drill bit compared to the rolling cutter. The hybrid design consistently comes in-gage without any debris left in hole. At the same time, the hybrid bit manages to drill stringers better than the rolling cutter. Additionally, when the hybrid design is given the chance, it reduced the number of bits utilized and it drilled the longest footage in sandstone lithology.

Area of Interest B

In this area, the same sandstone lithology is drilled laterally. In this well, the rolling cutter bit was being used as the primary type of bits. The bit was effectively drilling the lateral with 1184 and 874 ft per unit deployed. The interval per bit then dropped to 475 ft with slightly higher wear. The decision was made to try drilling using MRSS BHA which reduced the footage even further to 217 ft, then switched back to RSS again in which it gave a slightly improvement in footage to 237 ft. Both bits came out heavily damaged with even more sever dull for the bit that drilled on MRSS.

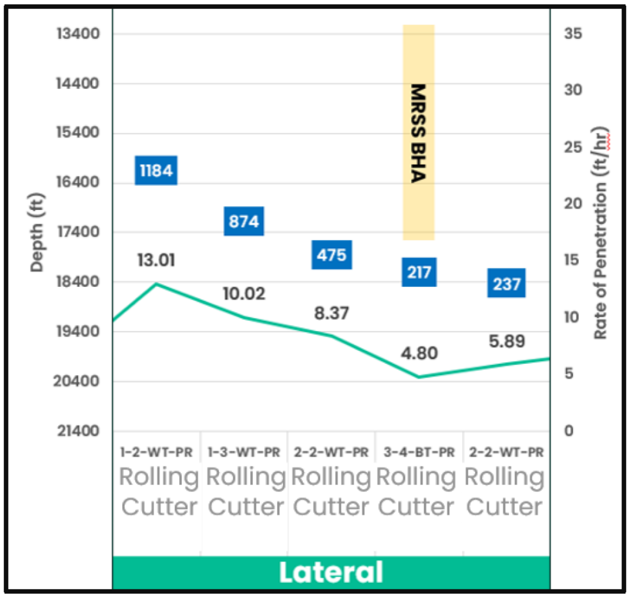

Figure 16—Area of Interest B Rolling Cutter Runs

In hope to improve footage drilled per bit, the decision was made to move out of typical choice and try alternative solutions. Three hybrid drill bits were utilized to reach section TD. The bits drilled 805 ft, 815 ft and the remaining 469 ft in sandstone lithology to finish the section. This represents 270% improvement in footage drilled compared to the last 2 runs drilled by the Rolling Cutter drill bit. Although the dull was heavy for these three deployments, the bits were still in-gage and left no debris or junk in hole. The hybrid design performance exceeded expectation. Not only did it save cost on the number of assets utilized and tripping time, but it also saved excessive non-productive time that could be required to ream the hole.

To make sure this is not just mere luck or a change in the lithology, further analysis was conducted to verify the performance. Looking at Figure 17, the rolling cutter bit that drilled 1184 ft went through homogenous sandstone lithology. The second one that drilled 874 ft went through 100% sand lithology for almost half of the run then had to drill 50% siltstone. A clear trend can be seen of decreasing ROP and footage drilled as the percent of siltstone increases when drilling with the Rolling Cutter bit. On the other hand, the Hybrid drill bit went through the same percentage or higher of siltstone and still managed to drill 4 times the footage compared to the last two runs that were drilled by the Rolling Cutter bit.

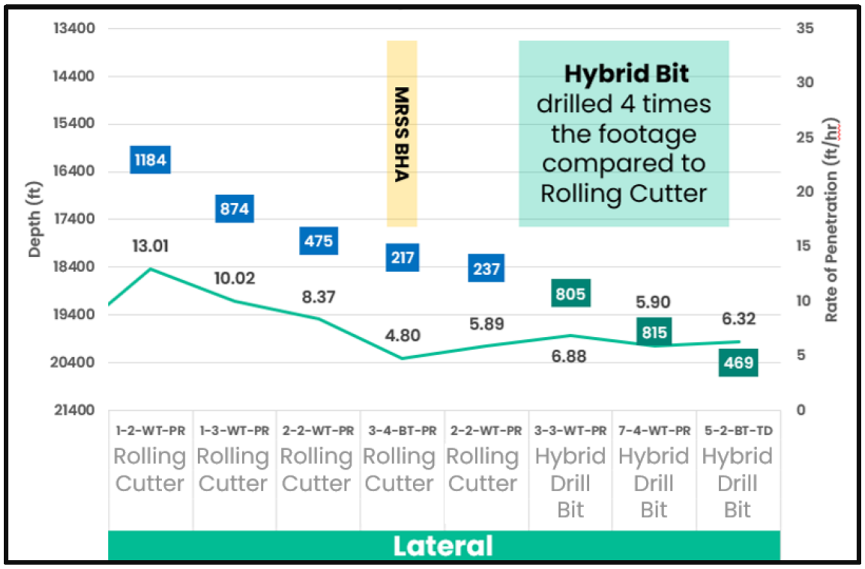

Figure 17—Area of Interest B Hybrid Drill Bit Runs

Intervals with similar siltstone percentage were compared to analyze the difference in performance between the Rolling Cutter bit and the Hybrid design. As can be seen from Figure 18, the Hybrid bit drilled the same percentage and interval of siltstone compared to the Rolling Cutter bit and yet continued to drill almost 4 times the footage. The hybrid drill bit showed higher ROP in siltstone with the same parameters applied compared to the Rolling Cutter.

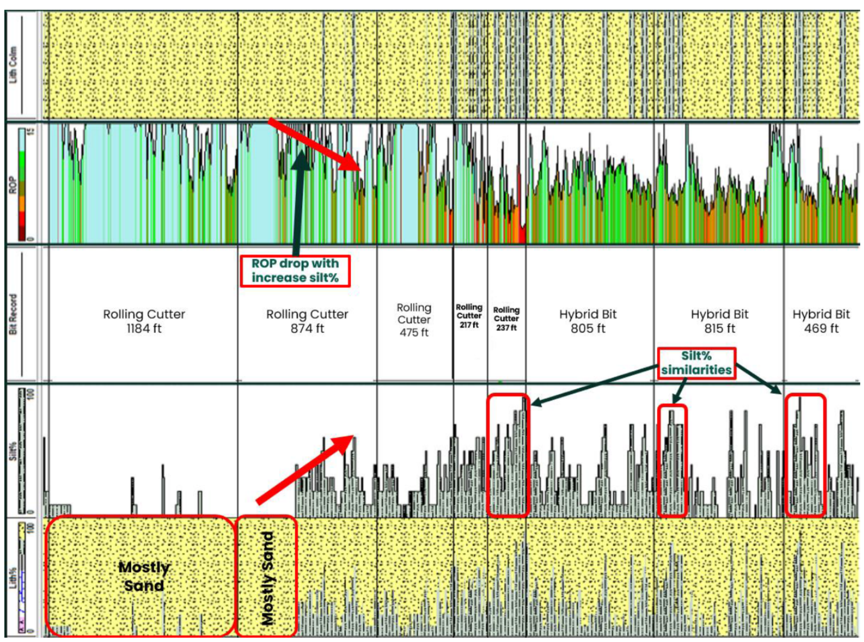

Figure 18—Area of Interest B Lithology Comparison

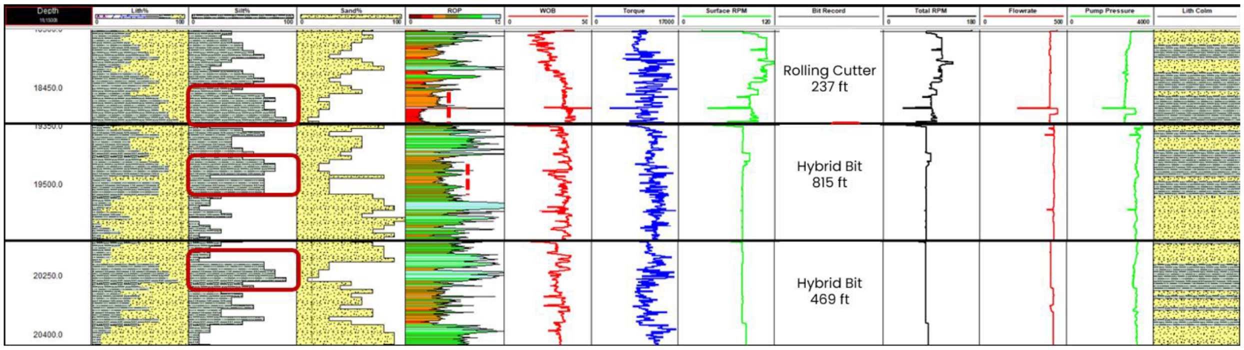

Figure 19—Area of Interest B: Comparing Performance in Similar Siltstone Percentage

The hybrid design showed higher performance than the rolling cutter bit in other interbedded carbonate lithology, as well. In the same well that was drilled in the curve section, the hybrid drill bit consistently drilled longer footage, run after run, and eventually landed the well.

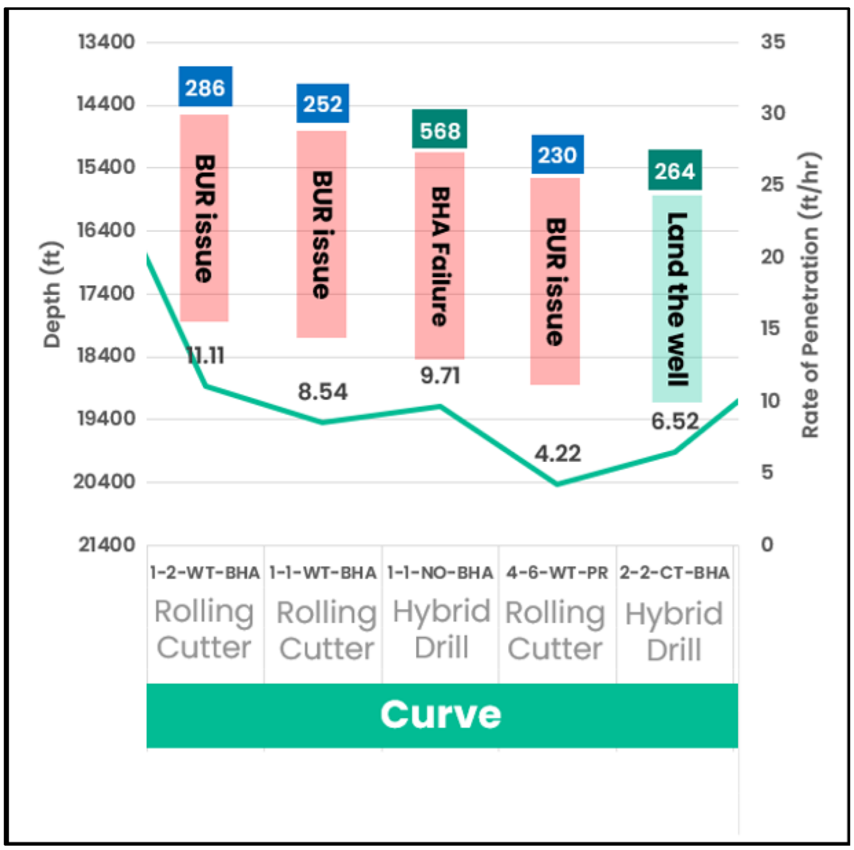

Figure 20—Area of Interest B Curve section

Other papers have reported similar results of extended footage and higher performance in interbedded sandstone lithology when comparing hybrid drill bits with fixed cutter and roller cone bits. The following table summarizes findings from 3 different papers that show gain in performance with hybrid technology in sandstone and other interbedded lithology.

Table 1—Summary of Papers that Compared Hybrid Technology to Rolling Cutter Drill Bit

Paper Title	Key Performance Metrics	Outcome Summary	Authors	Year
Hybrid Bits Offer Distinct Advantages in Selected Roller-Cone and PDC-Bit Applications	50% reduction in torsional oscillations, improved toolface control	Hybrid bits outperformed PDC bits with rolling cutters in interbedded formations.	Rolf Pessier & Michael Damschen	2011
A Breakthrough Performance for an Inland Application with a Hybrid Bit Technology	100% farther drilled, 2× ROP	Hybrid bits maintained consistent performance and reduced bit trips in hard stringers.	M. Di Pasquale, E. Calvaresi, S. Pecantet	2013
Hybrid Drill Bit Improves Drilling Performance in Heterogeneous Formations in Brazil	44% reduction in drilling time, doubled run length	Hybrid bits achieved significant performance gains in abrasive sandstone and shale.	I. Thomson, R. Krasuk, N. Silva, K. Romero	2011

Conclusion

In conclusion, the deployment of hybrid drill bits combining roller cones and PDC cutters has demonstrated significant advantages over conventional PDC bits with rotating cutters in abrasive, interbedded sandstone formations. The trials conducted in both area of interests A and B have shown that hybrid bits consistently outperform their counterparts in terms of footage drilled, rate of penetration (ROP), and overall bit durability.

In Area of Interest A, the hybrid bits were able to drill through harder and more interbedded lithologies, maintaining higher ROP and reducing the number of bits required to reach total depth (TD). The hybrid bits exhibited better wear resistance, coming out in-gage and with minimal cutter breakage, which significantly reduced non-productive time (NPT) associated with reaming and junk removal. The ability of hybrid bits to handle high UCS stingers and abrasive sandstone intervals with fewer trips and higher efficiency highlights their robustness and adaptability in challenging drilling environments.

Similarly, in Area of Interest B, the hybrid bits achieved remarkable improvements in footage drilled per bit, even in highly abrasive and interbedded sandstone formations. The comparison of lithology percentages and UCS values between hybrid bits and rolling cutter PDC bits revealed that hybrid bits consistently drilled longer intervals with higher ROP, despite encountering similar or more challenging formation conditions. The hybrid bits' performance in maintaining borehole quality and reducing the need for side-track jobs further shows their operational benefits.

The introduction of hybrid drill bit technology has proven to be a game-changer in hard rock drilling applications. By combining the shearing action of PDC cutters with the crushing action of roller cones, hybrid bits offer a more efficient solution for drilling through complex lithologies. The field data and post-run evaluations presented in this paper provide compelling evidence of the hybrid bits' superiority in terms of cost savings, time efficiency, and overall drilling performance.

As the industry continues to seek innovative solutions to enhance drilling operations, the adoption of hybrid drill bits will play a crucial role in overcoming the limitations of conventional bit designs. The successful deployment of hybrid bits in abrasive sandstone formations sets a benchmark over other competing drill bits genre. Future research and field trials will further refine hybrid bit technology, optimizing its design and performance to meet the evolving needs of the oil and gas industry.

In summary, the hybrid drill bits have demonstrated their capability to drill faster, longer, and more efficiently in abrasive, interbedded sandstone formations. The operational benefits, including reduced bit consumption, minimized NPT, and improved borehole quality, make hybrid bits a valuable addition to the drilling toolkit. The findings of this paper emphasize the importance of embracing hybrid technology to achieve better drilling outcomes and drive cost-effective operations in the pursuit of hydrocarbons.

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